

# PR ORIGIN GLOBAL

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GLOBAL EXPORT DIVISION

HIGH-PERFORMANCE  
FILTRATION MEDIA

## COCONUT SHELL ACTIVATED CARBON

*"Engineered from Nature.  
Perfected by Steam."*

WATER PURIFICATION

AIR FILTRATION



# WHAT IS COCONUT SHELL ACTIVATED CARBON?

Coconut shell activated carbon (CSAC) is a high-purity carbon adsorbent produced from coconut shells - a renewable biomass resource. Through rigorous carbonization and advanced steam activation, the raw shell is converted into a highly porous material capable of adsorbing contaminants at the molecular level with unmatched efficiency.

Unlike coal or wood-based alternatives, coconut shell carbon offers a unique microporous structure that makes it the premier choice for critical purification applications across heavy industries, municipal plants, and mining sectors.

## Why Coconut Shell?

- [x] High Carbon Content:**  
Naturally dense raw material yields superior fixed carbon.
- [x] Exceptional Hardness:**  
High resistance to abrasion & friction during backwash.
- [x] Natural Microporosity:**  
Tightly packed pore structure traps small molecules (VOCs).
- [x] Low Ash Formation:**  
Significantly cleaner than coal carbons, protecting systems.
- [x] 100% Renewable:**  
Ethically sourced from mature tropical coconut plantations.

## The PR Origin Advantage

- By controlling the supply chain from raw shell procurement to high-temp steam activation, we ensure consistency across every batch.
- [v] Strict Quality Control:**  
Laboratory tested to international ASTM standards.
  - [v] Custom Sizing:**  
Precise mesh screening minimizes pressure drop in filters.
  - [v] Global Logistics:**  
ISPM-15 compliant palletization & moisture-proof export.

## MANUFACTURING PROCESS

Our coconut shell activated carbon is produced through a meticulously controlled two-stage thermal process, ensuring absolute purity and structural maximization.

### 1. Carbonization

Mature coconut shells are heated in a controlled, oxygen-limited rotary kiln to remove volatile matter and moisture, converting raw biomass into dense, high-carbon char.



### 2. Steam Activation

The char is subjected to high-temperature steam (physical activation) at 900 deg C - 1100 deg C. This selective oxidation develops an intricate network of micropores, maximizing surface area.

### THE RESULT

## Surface Area: 900 - 1300 m<sup>2</sup>/g

A single gram of our activated carbon has the internal surface area equivalent to several tennis courts, creating a massive matrix for molecular adsorption and high-capacity purification.

# TECHNICAL SPECIFICATIONS

Manufactured to strict export tolerances, our CSAC meets the performance standards required by global municipal and industrial operators.

| PARAMETER          | TYPICAL RANGE / GUARANTEED       | TESTING METHOD       |
|--------------------|----------------------------------|----------------------|
| Iodine Value       | 900 - 1200 mg/g                  | ASTM D4607           |
| Surface Area (BET) | 900 - 1300 m2/g                  | Nitrogen Adsorption  |
| Moisture Content   | <= 5%                            | ASTM D2867           |
| Ash Content        | <= 5% (<= 3% for premium grades) | ASTM D2866           |
| Bulk Density       | 450 - 550 kg/m3                  | ASTM D2854           |
| Hardness           | > 95% (>= 97% for gold-recovery) | ASTM D3802           |
| Porosity Structure | Predominantly Microporous        | Pore Volume Analysis |
| pH                 | 7 - 11                           | ASTM D3838           |

## Quality Assurance

Every batch is rigorously tested in our laboratory prior to dispatch. Certificate of Analysis (COA) is provided with every commercial export shipment to guarantee compliance with your technical requirements.

# AVAILABLE GRADES & FORMATS

We supply activated carbon engineered to the specific hydrodynamic and aerodynamic requirements of your filtration systems.

|                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                     |                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div>Granular (GAC)</div><div><p>Irregularly shaped granules perfect for liquid phase filtration, offering high flow rates.</p></div><div><div>[x] 4 x 8 mesh</div><div>[x] 6 x 12 mesh</div><div>[x] 8 x 30 mesh</div><div>[x] 12 x 40 mesh</div></div></div> | <div><div>Powdered (PAC)</div><div><p>Pulverized carbon providing rapid adsorption kinetics for batch process units.</p></div><div><div>[x] 80 mesh</div><div>[x] 200 mesh</div><div>[x] 325 mesh</div></div></div> | <div><div>Pelletized</div><div><p>Extruded cylindrical pellets designed for low pressure drop in vapor phase systems.</p></div><div><div>[x] 1.5 mm diameter</div><div>[x] 3.0 mm diameter</div><div>[x] 4.0 mm diameter</div></div></div> |
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# GLOBAL INDUSTRY APPLICATIONS

## 1. Water Purification & Treatment

Crucial for drinking water treatment, industrial water filtration, RO pre-treatment, groundwater purification, and municipal water plants.

- [x] Removes chlorine & chloramines
- [x] Removes taste & odor compounds
- [x] Eliminates organic contaminants
- [x] Filters trace pesticides & VOCs

## 2. Air & Gas Filtration

Widely implemented in commercial HVAC, industrial exhaust scrubbing, and vapor recovery units.

- [x] VOC (Volatile Organic Compounds) capture
- [x] Industrial solvent vapor recovery
- [x] Toxic gas abatement and flue scrubber beds
- [x] Odor control in wastewater facilities

## 3. Gold Recovery & Metallurgy

High-hardness CSAC is the premier medium for Carbon-in-Leach (CIL) and Carbon-in-Pulp (CIP) circuits.

- [x] Abrasion resistance > 97% minimizes gold loss
- [x] High gold adsorption capacity (K-value)
- [x] Plateau loading kinetics for optimum elution
- [x] Low plate-shaped particle formation

4. Food & Beverage

Ensures absolute purity and clarity through beverage filtration, sugar decolorization, edible oil purification, and alcohol/spirit processing.

5. Pharma & Chemical

Highly pure grades used for critical solvent recovery, chemical catalysis, and Active Pharmaceutical Ingredient (API) purification processes.

COCONUT SHELL VS. COAL-BASED

When compared to traditional coal-based carbon, PR Origin Global's Coconut Shell Activated Carbon delivers superior mechanical and adsorptive properties, directly impacting your operational ROI.

| PERFORMANCE METRIC | OUR COCONUT SHELL         | STANDARD COAL-BASED         |
|--------------------|---------------------------|-----------------------------|
| Material Hardness  | > 95% (Ultra-durable)     | ~ 70% - 80% (Prone to dust) |
| Ash Content        | <= 5% (High purity)       | 8% - 15% (Higher residue)   |
| Surface Area (BET) | Up to 1300 m2/g           | 800 - 1000 m2/g             |
| Pore Structure     | Predominantly Microporous | Meso/Macroporous            |
| Eco-Footprint      | 100% Renewable Biomass    | Mined Fossil Fuel           |

# PACKAGING & STORAGE

To ensure our high-iodine activated carbon reaches your facility in pristine, moisture-free condition, we employ rigorous industrial packaging standards tailored for global maritime export.

## Standard Export Packaging

- [x] 25 kg Bags:**  
Multi-wall Kraft paper bags or heavy-duty PP woven bags with inner PE moisture-proof liners.
- [x] Jumbo Bags:**  
500 kg or 1000 kg FIBC bulk bags with inner liners and bottom discharge spouts for automated hopper feeding.
- [x] Palletization:**  
Shrink-wrapped and secured on ISPM-15 compliant heat-treated export pallets.

## Custom Requirements

Private labeling, OEM packaging, and custom weight bagging are available for wholesale distributors and specific batch-dosing requirements.

## Critical Storage & Safety

Store in a dry, well-ventilated area away from direct sunlight. Activated carbon is a strong reducing agent. Keep strictly away from strong oxidizers to prevent spontaneous combustion.

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## READY TO UPGRADE YOUR FILTRATION?

Contact our Global Export Division to discuss your specific mesh size requirements, request a commercial sample, or secure forward-contract pricing for CIF/FOB delivery.

### Export & Sales Inquiry

**Executive Contact: Yogesh Patel**

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"Engineered  
for  
Performance.

Trusted  
for  
Reliability."

ISO  
9001:2015  
CERTIFIED  
EXPORTER